SAI ABHINAV PATTOOR

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SUMMARY

Data Engineer-focused Computer Science senior graduating May 2027 (3.95 GPA). Automate production data workflows at Texas Tech, reconciling 50,000+ records weekly and saving 5+ hours per week with Python validation scripts. Project work includes scheduled AWS ETL pipelines, source-linked retrieval and MCP tools, and Node.js backends. Seeking new-grad Data Engineer and Software Engineer roles.

EDUCATION

Texas Tech University | B.S. Computer Science, Minor in Mathematics Expected May 2027

GPA 3.95/4.00 | President's Honor List: Fall 2025, Spring 2026

EXPERIENCE

Texas Tech University, Enrollment Management IT Lubbock, TX

Student Assistant, Data & Systems Engineering Feb 2026 - Present

- Reconcile **50,000+ records weekly** across Informatica ETL loads, using source-to-target checks to catch rejected or missing records.
- Automated vendor-file validation and cleanup in Python, saving **5+ hours weekly**; added regression tests for imports and record-resolution logic.
- Built a Python duplicate-resolution tool using deterministic matching and survivorship rules, routing ambiguous student records for human review.
- Develop and test Apex record validation and synchronization before production deployment.

Texas Tech University, Enrollment Management IT Lubbock, TX

IT Systems Intern, Automation & Data Quality May - Aug 2026

- Replaced routing formulas with a staff-editable lookup for **100+ programs**, removing the need for formula changes.
- Stored reporting values with Salesforce Flow and enforced unique IDs to prevent duplicate imports.

ProcureX Technologies Hyderabad, India

Technical Systems & Data Engineering Intern Jun - Aug 2025

- Caught **15 part-ID and unit-of-measure mismatches** before integration; documented each for engineering review.
- Checked **40+ procurement data flows** across **8 components** using Python and Excel.

PROJECTS

Austin Real Estate Analyzer | Python, Prefect, SQLite, AWS Dec 2025 - Nov 2026

- Built a Streamlit dashboard comparing home values and historical changes across **42 Austin ZIP codes**.
- Added filters and transparent scores based on home values and 1- and 3-year changes.
- Scheduled weekly AWS refreshes of monthly housing data, with validated snapshots for recovery after restarts.

ZipRAG | Python, SQLite, MCP Aug 2026

- Built an app that finds Austin neighborhood references and ZIP-code housing scores, returning source links so results can be checked.
- Retrieved the expected source in the top 3 on **all 10 development questions**; exposed search and score lookup as MCP tools for compatible assistants.

The Prospector | Node.js, Express.js, MongoDB Sep 2025

- Co-developed a **HackWesTX runner-up** search app; deployed a demo combining GitHub and Hacker News results with favorites and search history.
- Built the Express backend and MongoDB saved-search schema; removed duplicate URLs and handled source failures with request timeouts.

LEADERSHIP

Marketing Lead | Google Developer Groups at Texas Tech Aug - Dec 2025

- Led chapter marketing and helped host technical workshops for student developers.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript
Data & Backend: pandas, Informatica, Prefect, Node.js/Express, REST APIs, SQLite, MongoDB
Cloud & Tools: AWS (S3, CodeBuild, EventBridge, CloudFront, CloudFormation), Git/GitHub
Frameworks & Platforms: React, Streamlit, scikit-learn, MCP, Salesforce (Apex, Flow, SOQL)